SOURCE CODE:

```
def encryptDecrypt(inpString):
    xorKey ='H';
    length = len (inpString);
    for i in range (length):
        inpString = (inpString[:i] + chr(ord(inpString[i]) ^ ord (xorKey)) +
inpString[i + 1:]);
        print(inpString[i], end = "");
    return inpString;
if __name__ == '__main__':
    sampleString = "hello world";
    print("encrypted string:", end = "");
    sampleString = encryptDecrypt(sampleString);
    print("\n")


    print("decrypted string: ", end = "");
    encryptDecrypt(sampleString);
```

OUTPUT:

```
encrypted string: -$$'h?':$,

decrypted string: hello world
```

SOURCE CODE:

```
def encrypt(text,s):
    result=""

    for i in range(len(text)):
        char=text[i]

        if (char.isupper()):
            result+=chr((ord(char)+s-65)%26+65)

        else:
            result+=chr((ord(char)+s-97)%26+97)

    return result

text=input("Enter the plain text to convert ceaser cipher:")
s=5
print("Text:"+text)
print("Shift:"+str(s))
print("Cipher:"+encrypt(text,s))
```

```
def decrypt(text,s):
    result=""
    for i in range(len(text)):
        char=text[i]
        if(char.isupper()):
            result+=chr((ord(char)+s-65)%26+65)
        else:
            result+=chr((ord(char)+s-97)%26+97)
    return result
text=input("Enter the ceaser cipher to convert plain text:")
s=26-5
print("Texet:"+text)
print("Shift:"+str(s))
print("Cipher:"+decrypt(text,s))
```

OUTPUT:

```
Enter the plain text to convert ceaser cipher:beautifulworld
Text:beautifulworld
Shift:5
Cipher:gjfzynkzqbtwqi
Enter the ceaser cipher to convert plain text:gjfzynkzqbtwqi
Texet:gjfzynkzqbtwqi
Shift:21
Cipher:beautifulworld
```

SOURCE CODE:

```
keyMatrix=[[0] * 3 for i in range(3)]
messageVector=[[0] for i in range(3)]
cipherMatrix=[[0] for i in range(3)]
def getKeyMatrix(key):
    k=0
    for i in range(3):
        for j in range(3):
            keyMatrix[i][j] = ord(key[k])  %  65
            k+=1

def encrypt(messageVector):
    for i in range(3):
        for j in range(1):
            cipherMatrix[i][j]=0
            for x in range(3):
                cipherMatrix[i][j] += (keyMatrix[i][x] * messageVector[x][j])
            cipherMatrix[i][j] = cipherMatrix[i][j] % 26

def HillCipher(message,key):
    getKeyMatrix(key)
    for i in range(3):
```

```
        messageVector[i][0]=ord(message[i]) % 65
    encrypt(messageVector)
    CipherText=[]
    for i in range(3):
        CipherText.append(chr(cipherMatrix[i][0]+65))
    print("Ciphertext:","".join(CipherText))

def main():
    message=str(input("Enter the plain text:"))
    key=str(input("Enter the key:"))
     HillCipher(message,key)
if __name__=="__main__":
    main()
```

OUTPUT:

```
Enter the plain text:human
Enter the key:allareequal
Ciphertext: UYA
```

SOURCE CODE:

```
import string

all_letters=string.ascii_letters

"""create a dictionary to store the substitution for the given  alphabet in
the plaintext based on the key"""

dict1 = {}

key = 4

for i in range(len(all_letters)):

    dict1[all_letters[i]]=all_letters[(i+key)%len(all_letters)]

plain_txt=str(input("Enter your plain_txt:"))

cipher_txt=[]

for char in plain_txt:

    if char in all_letters:

        temp=dict1[char]

        cipher_txt.append(temp)

    else:

        temp=char

        cipher_txt.append(temp)

cipher_txt="".join(cipher_txt)

print("cipher text is:",cipher_txt)

"""create a dictionary to store the substitution for the given alphabet in
the cipher based on the key"""

dict2 = {}
```

```
for i in range(len(all_letters)):
    dict2[all_letters[i]]=all_letters[(i-key) % (len(all_letters))]
decrypt_txt=[]
for char in cipher_txt:
    if char in all_letters:
        temp=dict2[char]
        decrypt_txt.append(temp)
    else:
        temp=char
        decrypt_txt.append(temp)
decrypt_txt="".join(decrypt_txt)
print("Recovered plain text:",decrypt_txt)
```

OUTPUT:

```
Enter your plain_txt:intelligent
cipher text is: mrxippmkirx
Recovered plain text: intelligent
```

SOURCE CODE:

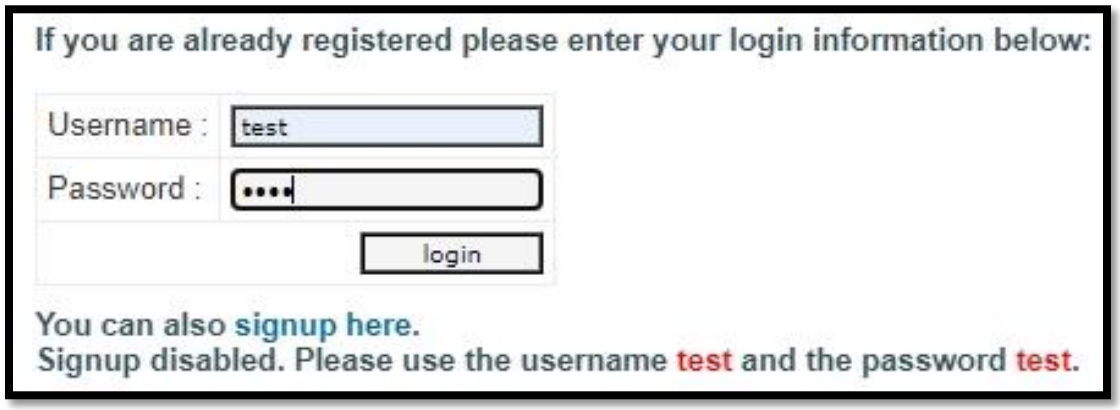
```
from Crypto.Cipher import DES


def pad(text):
    # add padding to the input text to ensure it's a multiple of 8 bytes
    while len(text) % 8 != 0:
        text += b' '
    return text


def encrypt(key, plaintext):
    cipher = DES.new(key, DES.MODE_ECB)
    padded_text = pad(plaintext)
    ciphertext = cipher.encrypt(padded_text)
    return ciphertext


def decrypt(key, ciphertext):
    cipher = DES.new(key, DES.MODE_ECB)
    padded_text = cipher.decrypt(ciphertext)
    plaintext = padded_text.rstrip(b' ')
    return plaintext


# example usage
```

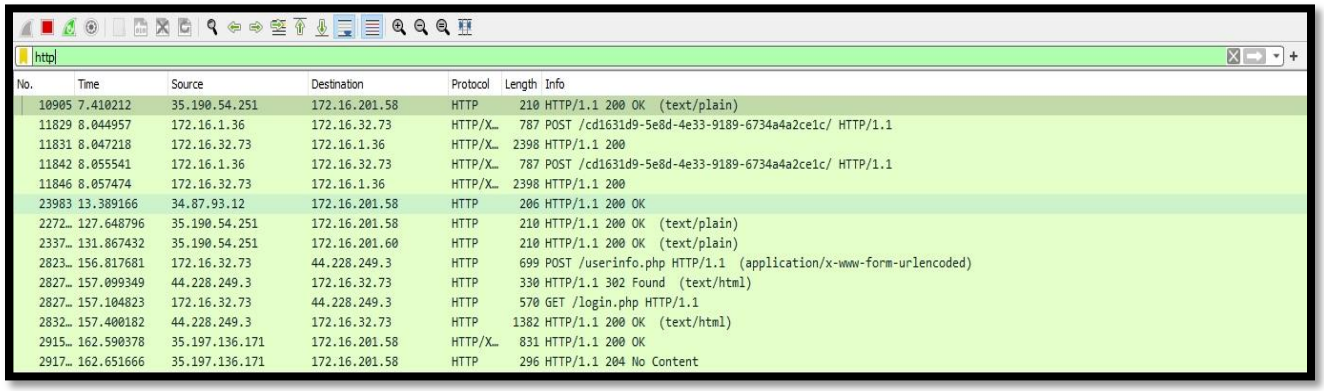
```
key = b'secretke'
plaintext = b'This is a secret message.'
ciphertext = encrypt(key, plaintext)
decrypted_text = decrypt(key, ciphertext)

print('Plaintext:', plaintext)
print('Ciphertext:', ciphertext)
print('Decrypted text:', decrypted_text)
```

OUTPUT:

```
Plaintext: b'This is a secret message.'
Ciphertext: b'\xc1<l\xc57\x9d\xa4\xbd7\xd16\x8f(\x05\x13\xf6%I\xc5;\xd4\xd3\x96\xd2\xc4\xa2\xc6\xbdw\xa1:\\'
Decrypted text: b'This is a secret message.'
```

OUTPUT:



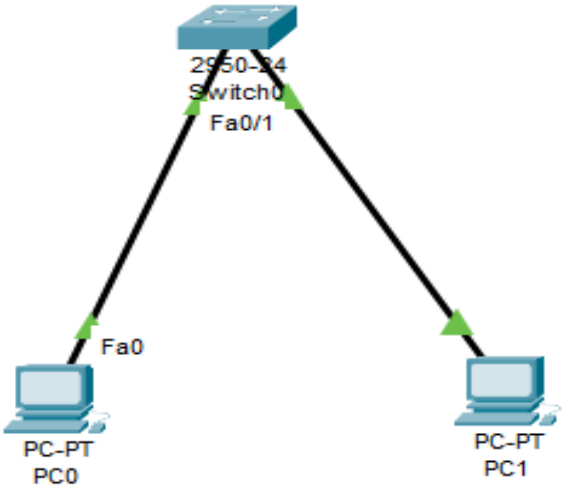


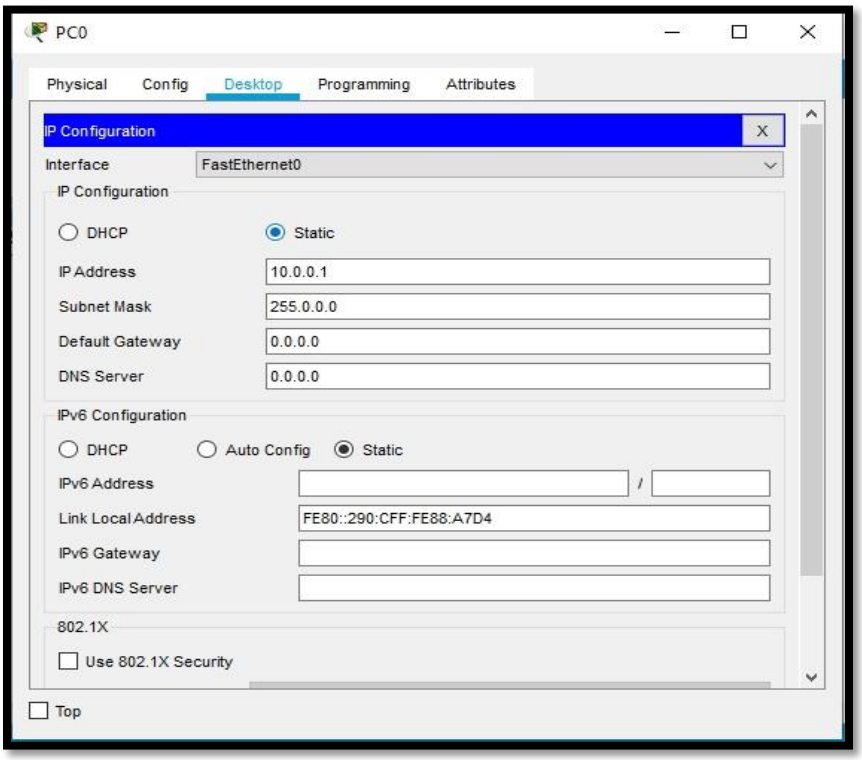
http

No.	Time	Source	Destination	Protocol	Length	Info
10905	7.410212	35.190.54.251	172.16.201.58	HTTP	210	HTTP/1.1 200 OK (text/plain)
11829	8.044957	172.16.1.36	172.16.32.73	HTTP/X…	787	POST /cd1631d9-5e8d-4e33-9189-6734a4a2ce1c/ HTTP/1.1
11831	8.047218	172.16.32.73	172.16.1.36	HTTP/X…	2398	HTTP/1.1 200
11842	8.055541	172.16.1.36	172.16.32.73	HTTP/X…	787	POST /cd1631d9-5e8d-4e33-9189-6734a4a2ce1c/ HTTP/1.1
11846	8.057474	172.16.32.73	172.16.1.36	HTTP/X…	2398	HTTP/1.1 200
23983	13.389166	34.87.93.12	172.16.201.58	HTTP	206	HTTP/1.1 200 OK
2272…	127.648796	35.190.54.251	172.16.201.58	HTTP	210	HTTP/1.1 200 OK (text/plain)
2337…	131.867432	35.190.54.251	172.16.201.60	HTTP	210	HTTP/1.1 200 OK (text/plain)
2823…	156.817681	172.16.32.73	44.228.249.3	HTTP	699	POST /userinfo.php HTTP/1.1 (application/x-www-form-urlencoded)
2827…	157.099349	44.228.249.3	172.16.32.73	HTTP	330	HTTP/1.1 302 Found (text/html)
2827…	157.104823	172.16.32.73	44.228.249.3	HTTP	570	GET /login.php HTTP/1.1
2832…	157.400182	44.228.249.3	172.16.32.73	HTTP	1382	HTTP/1.1 200 OK (text/html)
2915…	162.590378	35.197.136.171	172.16.201.58	HTTP/X…	831	HTTP/1.1 200 OK
2917…	162.651666	35.197.136.171	172.16.201.58	HTTP	296	HTTP/1.1 204 No Content

```
uname=test&pass=1234HTTP/1.1 302 Found
Server: nginx/1.19.0
Date: Thu, 06 Apr 2023 09:50:25 GMT
Content-Type: text/html; charset=UTF-8
Transfer-Encoding: chunked
Connection: keep-alive
X-Powered-By: PHP/5.6.40-38+ubuntu20.04.1+deb.sury.org+1
Location: login.php
```

OUTPUT:


2950-24
Switch0
Fa0/1
Fa0
PC-PT
PC0
PC-PT
PC1


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IP Address
10.0.0.1
Subnet Mask
255.0.0.0
Default Gateway
0.0.0.0
DNS Server
0.0.0.0
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::290:CFF:FE88:A7D4
IPv6 Gateway
IPv6 DNS Server
802.1X
Use 802.1X Security
Top

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Link-local IPv6 Address.........: FE80::290:CFF:FE88:A7D4
   IP Address......................: 10.0.0.1
   Subnet Mask.....................: 255.0.0.0
   Default Gateway.................: 0.0.0.0

Bluetooth Connection:

   Link-local IPv6 Address.........: ::
   IP Address......................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: 0.0.0.0

C:\>
```

PC1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

DHCP | Static

IP Address: 10.0.0.2

Subnet Mask: 255.0.0.0

Default Gateway: 0.0.0.0

DNS Server: 0.0.0.0

IPv6 Configuration

DHCP | Auto Config | Static

IPv6 Address: /

Link Local Address: FE80::2E0:A3FF:FE54:9D7A

IPv6 Gateway:

IPv6 DNS Server:

802.1X

Use 802.1X Security

Top

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Link-local IPv6 Address.........: FE80::2E0:A3FF:FE54:9D7A
   IP Address......................: 10.0.0.2
   Subnet Mask.....................: 255.0.0.0
   Default Gateway.................: 0.0.0.0

Bluetooth Connection:

   Link-local IPv6 Address.........: ::
   IP Address......................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: 0.0.0.0

C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- **Switch#enable**
- **Switch#show ip interface brief**
- **Switch#show port-security**
- **Switch#show port-security address**
- **Switch#show mac-address-table**

```
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.a3d0.2216    DYNAMIC     Fa0/10
   1    00d0.9773.de14    DYNAMIC     Fa0/1
```

- **Switch#configure terminal**
- **Switch(config)#interface fastEthernet 0/10**
- **Switch(config-if)#switchport mode access**
- **Switch(config-if)#switchport port-security**
- **Switch(config-if)#switchport port-security mac-address ?**
- **Switch(config-if)#switchport port-security mac-address sticky**
- **Switch(config-if)#switchport port-security maximum ?**
- **Switch(config-if)#switchport port-security maximum 1**
- **Switch(config-if)#switchport port-security violation ?**
- **Switch(config-if)#switchport port-security violation**
- **Switch(config-if)#switchport port-security violation shutdown**
- **Switch(config-if)#^Z**
- **Switch#show port-security**

- **Switch#show port-security address**
- **Go to PC2 command prompt and enter the below the command**
    - **ping 10.0.0.1**

```
Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128
Reply from 10.0.0.1: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

- **After go to the Switch CLI**
- **Switch#show port-security address**

```
                                Secure Mac Address Table
-------------------------------------------------------------------------------
Vlan       Mac Address Type                            Ports                  Remaining Age
                                                                                     (mins)
----       -----------  ----                           -----                  -------------
1          0030.A3D0.2216         SecureSticky                      FastEthernet0/10
-------------------------------------------------------------------------------
Total Addresses in System (excluding one mac per port)     : 0
Max Addresses limit in System (excluding one mac per port) : 1024
```

- **Switch#show port-security**

```
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
                (Count)       (Count)         (Count)
--------------------------------------------------------------------
       Fa0/10        1           1                 0          Shutdown
--------------------------------------------------------------------
```

SOURCE CODE:

```
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes

# Generate a new RSA key pair
private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)
public_key = private_key.public_key()

# Hash the message to be signed
message = b"This is a message to be signed"
hash_algorithm = hashes.SHA256()
hash_object = hashes.Hash(hash_algorithm)
hash_object.update(message)
digest = hash_object.finalize()

# Sign the hashed message using the private key
signature = private_key.sign(
    digest,
    padding.PKCS1v15(),
    hashes.SHA256()
)

# Verify the signature using the public key
try:
    public_key.verify(
        signature,
```

```
        digest,
        padding.PKCS1v15(),
        hashes.SHA256()
    )
    print("The signature is valid.")
except:
    print("The signature is not valid.")
```

OUTPUT:

The signature is valid.